

3.1.11 Design for manufacturing, maintenance, repair and disposal

Manufacture, repair, maintenance and disposal

Content	Potential links to maths and science
The need to modify designs to make them more efficient to manufacture, including: • reducing the number of manufacturing processes • how the choice of materials affects the use, care and disposal of products: advisory labelling to encourage responsible use and care of textile products • application of the six Rs of sustainability: reduce the quantity of materials, of toxic materials, of damaging materials and associated energy use, reuse components and parts, rethink by using eco friendly alternative materials, recycle materials and/or components into new products • maintenance: temporary and integral fixings, use of standardised parts, allowing for service and repair/ replacement of parts, ability to upgrade with software downloads, selection of fabrics and components that can be cared for without the need for special treatments, advisory labelling to encourage responsible washing and drying of textile products.	

Care and maintenance of products

Content	Potential links to maths and science
Students should be aware of, and able to explain, the different ways in which a product can be cared for and maintained, including care labels, their use and what they mean.	Understand the relationship between care recommendations and fibre/fabric properties, eg the thermoplastic nature of synthetic fibres and wash/iron temperatures, shrinkage of wool fibres when washed at high temperatures and with excessive mechanical action.